Where Does the Signal Live? Representation and Markov Structure in Neural Differential Cryptanalysis

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Abstract

Neural differential distinguishers can break more rounds of block ciphers than classical single-bit methods, yet two foundational questions remain open: where does the distinguishing signal reside in observable ciphertext data, and does this signal compose across rounds in the memoryless fashion assumed by classical differential analysis? We answer both through a controlled study on three cipher families—Speck32/64 (ARX), Simon32/64 (Feistel), and Present-64/80 (SPN)—using seven architectures under identical conditions. Saliency and attention analysis localize the signal to XOR-difference bits, multi-bit representations outperform word-level inputs by 29 pp, and all seven architectures agree within 2 pp—confirming that the signal is intrinsic to cipher structure, not model capacity. Transfer experiments then reveal that signal composition is family-dependent: SPN features compose hierarchically (99.1% cross-round transfer, $p < 10^{-11}$), while ARX and Feistel features are round-specific and *anti-compose*—a 5-round Speck32/64 model scores 45.5% on 3-round data ($p < 10^{-8}$), below random chance. VAE-based validation independently confirms that the cipher's differential propagation *is* Markovian (memory/Markov MSE ratio ≈ 1.0 across all rounds)—yet neural features still anti-compose, indicating that the networks learn round-specific representations that go beyond the Markovian differential structure. These results fill a gap identified by a recent survey of 66 peer-reviewed papers [8]: no prior work jointly characterizes signal representation and compositional structure across cipher families.

1 Introduction

Lightweight block ciphers protect billions of IoT sensors, RFID tags, and embedded controllers where silicon area and energy are scarce [3, 7]. The three dominant design paradigms—ARX (Add-Rotate-XOR), Feistel networks, and Substitution-Permutation Networks (SPN)—each make distinct structural trade-offs between diffusion speed, parallelism, and hardware cost. Whether one paradigm is systematically more vulnerable to a given class of attack than another is a question with direct implications for cipher selection in constrained environments.

Classical differential cryptanalysis [6] answers this by exhaustive search over differential trails, an approach that becomes intractable as round counts grow. Automated methods based on MILP [14] extend the reach of classical analysis, but they optimize within a fixed algebraic framework and cannot discover features outside the differential distribution table (DDT).

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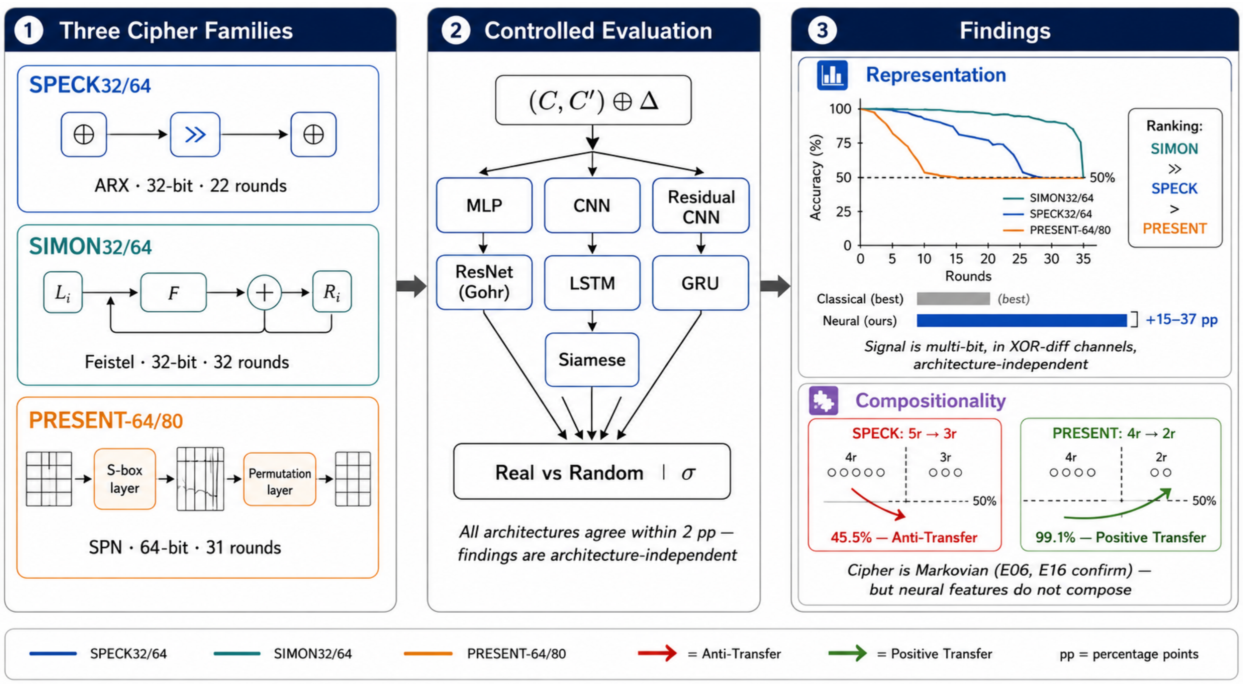

Figure 1: **Overview.** Identical neural distinguishers are trained on three cipher families (left) using seven architectures (center), yielding two sets of findings (right). *Representation:* signal resides in XOR-difference bits and is architecture-independent, establishing a vulnerability ordering SIMON32/64 $\gg$ SPECK32/64 $>$ PRESENT-64/80. *Compositionality:* SPN features compose hierarchically across rounds (positive transfer >99%), while ARX and Feistel features anti-compose (<50%), violating the Markovian assumption that underpins classical differential cryptanalysis.

In 2019, Gohr [9] demonstrated that a residual neural network, trained as a binary classifier on ciphertext pairs, could distinguish 8-round SPECK32/64 from random—surpassing the best known classical distinguisher. Since then, the field has expanded rapidly: a recent survey [8] catalogs 66 peer-reviewed papers training neural differential distinguishers across 24 primitives with 15+ architectures. Yet despite this breadth, two foundational questions remain unanswered.

The first is a question of **representation**: where does the distinguishing signal actually reside in observable ciphertext data? Prior work demonstrates that neural distinguishers outperform classical methods, but does not characterize *where* the signal lives—which bits, which input encoding, at what round depth—or whether the signal is a property of the model or of the cipher itself.

The second is a question of **compositionality**: does the learned signal obey the Markovian assumption that underpins classical differential analysis? If neural distinguishers learn the same kind of features as classical methods, their predictions should transfer hierarchically across round counts—a model trained at r rounds should also succeed at $r' < r$ rounds. Violations of this assumption, particularly *anti-transfer* where accuracy drops *below* 50%, would reveal that neural features are fundamentally non-composable and round-specific.

Our key insight is that *transfer polarity*—whether a distinguisher trained at one round count helps or hurts at another—serves as a direct empirical test of Markovian composability. Gohr et al. [10] proved that SPN and Feistel distinguishers can only learn differential features under independent round keys, while ARX distinguishers learn additional non-differential features. If SPN features are purely differential, they should compose hierarchically (positive transfer); if ARX features include non-differential components, they may be round-specific (anti-transfer). Our experiments confirm exactly this prediction and extend it to Feistel ciphers, where we observe unexpected anti-transfer.

We make the following contributions, organized around these two questions:

1. **Signal localization.**
   - A cross-family vulnerability ordering: SIMON32/64 is distinguishable 2–3 rounds deeper than SPECK32/64 or PRESENT-64/80 (Section 5.2.1), with all seven architectures agreeing within 2 pp(Section 5.2.3).
   - Saliency and attention analysis identifying the specific XOR-difference bits that carry the signal (Section 6.1).
   - A 15–37 ppneural advantage over single-bit classical baselines, confirming multi-bit signal invisible to classical analysis (Section 5.2.2).

2. **Compositional structure.**
   - Transfer polarity is family-dependent: SPECK32/64 (ARX) and SIMON32/64 (Feistel) exhibit *anti-transfer* ($\leq$48.6%, $p < 10^{-5}$), while PRESENT-64/80 (SPN) exhibits *positive transfer* ($\geq$96.9%, $p < 10^{-6}$) (Section 5.3.1). This is the first demonstration of below-chance transfer in neural cryptanalysis.
   - VAE-based Markov validation confirms that the cipher's differential propagation *is* memoryless (ratio ≈ 1.0 at all rounds)—yet neural features still anti-compose, demonstrating that the networks learn round-specific structure beyond the Markovian differential (Section C).
   - Key recovery fails on SIMON32/64 despite 100% distinguisher accuracy, revealing that non-composable signal cannot support Bayesian subkey search (Section 5.3.2).

Section 2 surveys related work. Section 3 formalizes the threat model and notation. Section 4 describes the experimental methodology. Section 5 presents the main results, organized by these two questions. Section 6 interprets the findings. Section 7 concludes.

2 Related Work

Classical differential cryptanalysis. Biham and Shamir [6] introduced differential cryptanalysis, tracing how input differences propagate through iterated block ciphers. Matsui [13] complemented this with linear cryptanalysis, exploiting affine approximations of round functions. Automated trail search using MILP [14] and SAT/SMT solvers has extended classical analysis to high round counts, but these methods operate within the DDT framework and cannot capture features outside it.

Neural differential distinguishers. Gohr [9] trained a depth-10 residual network to distinguish 8-round Speck32/64 from random at 51.4%, achieving 11-round key recovery via Bayesian scoring. Subsequent work has explored architecture design (Bellini et al.'s cipher-agnostic DBitNet [4]; attention mechanisms [12]; gated linear units [11]; SENet [2]), data scaling (Zhang et al. [16] use multi-pair inputs), and cipher breadth (24 primitives cataloged in [8]). The current state of the art on our target ciphers is: Simon32/64 at 11 rounds via DBitNet/SENet [4, 15], Speck32/64 at 8 rounds via Gohr's ResNet [9], and Present-64/80 at 9 rounds via DBitNet [4]. Our MLP-based results are intentionally weaker (fewer rounds) because our contribution is the controlled *comparison*, not depth maximization.

Explainability of neural distinguishers. Benamira et al. [5] showed that Gohr's network learns differential-linear features corresponding to truncated differential patterns in the penultimate rounds. Gohr, Leander, and Neumann [10] proved that under independent round keys, SPN and Feistel distinguishers can only learn features equivalent to the DDT, while ARX distinguishers exploit additional non-differential information. This theoretical result directly predicts our transfer findings: differential-only features (SPN) should compose hierarchically, while non-differential features (ARX) are round-specific.

Transfer learning in adversarial ML. Transfer attacks are well-studied for adversarial examples: perturbations crafted on one model transfer to another [9]. In cryptanalysis, however, no prior work has measured whether neural distinguishers transfer across ciphers or round counts. The closest work is Baksi et al. [1], who trained distinguishers on multiple lightweight ciphers but did not test cross-application.

Gap. The survey by Gérault et al. [8] catalogs 66 peer-reviewed papers in neural differential cryptanalysis. None jointly addresses where the distinguishing signal resides in ciphertext data and whether that signal composes across rounds in a Markovian fashion. We fill this gap with a controlled benchmark that holds architecture, data volume, input representation, and training protocol constant while varying only the cipher and round count, enabling direct answers to both questions.

3 Problem Formulation

3.1 Threat Model

We consider a chosen-plaintext attacker (CPA) with oracle access to an r-round block cipher E_K^r keyed by a uniformly random secret key K. The attacker knows a fixed input difference ΔP and receives ciphertext pairs (C, C') produced from plaintext pairs $(P, P \oplus \Delta P)$. The attacker does not know the key and has no access to intermediate round states.

Success metric. Let $f : \{0,1\}^{2b} \to [0,1]$ be a binary classifier that takes a ciphertext pair and outputs the probability that it was generated from the real distribution (label 1) versus the random distribution (label 0). We define the *distinguishing advantage* as

$$\varepsilon = \mathrm{acc}(f) - 0.5, \tag{1}$$

where $\mathrm{acc}(f)$ is the classification accuracy on a balanced test set. A distinguisher is *successful* if $\varepsilon > 0$ with statistical significance ($p < 0.05$ under a one-sample t-test against 0.5).

Classification in the n-m-T-E framework. Following the taxonomy of Bellini et al. [4] and Gérault et al. [8], our setup is **2-1-CT-R**: $n = 2$ ciphertext inputs, $m = 1$ output (binary label), ciphertext-only input type ($T = \mathrm{CT}$), and Gohr's "real vs. random" labeling ($E = \mathrm{R}$).

3.2 Target Ciphers

We study three lightweight block ciphers, each representing a major design paradigm:

Table 1: Target cipher specifications.

Cipher	Family	Block	Key	Rounds	Input diff. ΔP
Speck32/64	ARX	32	64	22	`(0x0040, 0x0000)`
Simon32/64	Feistel	32	64	32	`(0x0000, 0x0001)`
Present-64/80	SPN	64	80	31	`0x0000000000000001`

Speck32/64 uses modular addition, bitwise rotation, and XOR. Simon32/64 uses a balanced Feistel structure with AND, rotation, and XOR. Present-64/80 uses a 4-bit S-box layer followed by a bit permutation. All input differences follow Gohr's convention [9] or the natural single-bit choice for the respective cipher.

3.3 Transfer Protocol

To measure transfer, we train a distinguisher $f^{(c,r)}$ on cipher c at round r and evaluate it on data from:

- **Cross-round:** Same cipher, different round count $r' \neq r$.
- **Cross-cipher:** Different cipher $c' \neq c$, same representation.

We define *anti-transfer* as accuracy significantly below 50% ($p < 0.05$) and *positive transfer* as accuracy significantly above 50%. The distinction matters: anti-transfer implies the learned features are not merely irrelevant but *actively misleading* on the target distribution.

4 Methodology

4.1 Overview

Our experimental pipeline has four stages: data generation, representation, training, and evaluation. We deliberately fix all hyperparameters across cipher families so that any performance difference is attributable to cipher structure, not tuning.

4.2 Data Generation

For each cipher at r rounds, we generate a balanced dataset of $N = 500{,}000$ ciphertext pairs. Positive samples (label 1) encrypt $(P, P \oplus \Delta P)$ under a uniformly random key K; negative samples (label 0) encrypt two independently random plaintexts under the same K, following Gohr's protocol [9]. Each experiment uses 3–5 random seeds; the key changes with each seed.

4.3 Input Representation

We expand each ciphertext word into individual bits and concatenate:

$$\mathbf{x} = \mathrm{bits}(C_L) \| \mathrm{bits}(C_R) \| \mathrm{bits}(C'_L) \| \mathrm{bits}(C'_R) \| \mathrm{bits}(C_L \oplus C'_L) \| \mathrm{bits}(C_R \oplus C'_R), \tag{2}$$

yielding a 96-dimensional input for 32-bit block ciphers and 192-dimensional for Present-64/80 (64-bit). We denote this the *R2_xor_diff* representation.

Justification. Among six representations tested on Speck32/64 at 5 rounds (E02; Section A), R2_xor_diff achieves 88.7% accuracy, tied with the manually engineered R8_statistical (88.6%) and significantly above the word-level R5 baseline (59.8%). On Present-64/80 at 5 rounds, R2_xor_diff (76.1%) again ties with R8_statistical (76.1%) and dominates all others. On Simon32/64 at 7 rounds, R4_bit_sliced (85.6%) outperforms R2_xor_diff (81.3%) by 4.3 pp, suggesting that cipher-specific representation tuning could improve absolute accuracy. We retain R2_xor_diff across all families to preserve the controlled comparison that is our primary contribution.

4.4 Architecture

Our primary architecture is a feedforward MLP: four hidden layers of sizes $[256, 128, 64, 32]$ with batch normalization after each, followed by a single sigmoid output. This is Gohr's simpler MLP variant [9], deliberately chosen as a *lower bound* on neural distinguisher performance.

We also evaluate six additional architectures—CNN, Residual CNN, Gohr-MLP (larger), LSTM, GRU, and Siamese—on Speck32/64 at 5 rounds, as well as Gohr's depth-10 ResNet on all three ciphers (Section 5.2.3).

4.5 Training Protocol

All models use Adam ($\mathrm{lr} = 10^{-3}$), binary cross-entropy loss, batch size 5000, for up to 30 epochs with early stopping (patience 5) on validation accuracy. Training data is regenerated per seed. All experiments run on a single NVIDIA Tesla V100-PCIE-32GB.

4.6 Classical Baseline

As a classical reference, we implement a best-bit-bias distinguisher. For each of the $2b$ output difference bits, we compute $\epsilon_i = |p_i - 0.5|$ over 10^5 samples across 5 keys. The distinguisher classifies using the bit with maximum bias. This is a single-bit distinguisher, intentionally weaker than DDT-based methods [10]; we report it as a lower bound on classical performance and cite Gohr's published numbers for the stronger ResNet comparison.

5 Experiments and Results

5.1 Setup

We evaluate MLP distinguishers on three ciphers across the following round ranges: Speck32/64 (3–8), Simon32/64 (4–11), Present-64/80 (2–7). Each configuration uses 500K training samples, 3–5 random seeds, and the R2_xor_diff representation. Compute budget: approximately 48 GPU-hours total on a single NVIDIA V100-PCIE-32GB. Our experimental setting is 2-1-CT-R in the classification of Bellini et al. [4], Gérault et al. [8].

Baselines. We compare against (i) our best-bit-bias classical distinguisher, (ii) Gohr's published ResNet numbers on Speck32/64 (92.4% at 5r, 78.8% at 6r, 61.6% at 7r, 51.4% at 8r) [9], and (iii) our own ResNet trained under identical conditions.

5.2 Locating the Signal

We first characterize *where* the distinguishing signal resides: at what round depth, in which data dimensions, and whether it depends on the model or the cipher.

5.2.1 Signal Depth Across Cipher Families

Table 2: Neural distinguisher accuracy (%) vs. round count. Values are mean ± std over 3–5 seeds. 95% bootstrap CIs are reported for boundary rounds (accuracy < 55%). Dashes indicate untested configurations.

r	**Speck32/64 (ARX)**	**Simon32/64 (Feistel)**	**Present-64/80 (SPN)**
2	—	—	100.00 ± 0.00
3	99.99 ± 0.00	—	100.00 ± 0.00
4	97.78 ± 0.42	100.00 ± 0.00	98.39 ± 0.03
5	87.37 ± 0.49	99.97 ± 0.01	71.92 ± 0.23
6	66.96 ± 0.73	98.09 ± 0.08	52.55 ± 0.39
7	50.76 ± 0.10	79.09 ± 0.23	50.03 ± 0.02
8	49.98 ± 0.20	61.13 ± 0.42	—
9	—	52.56 ± 0.18	—
10	—	50.32 ± 0.23	—
11	—	50.02 ± 0.08	—

Table 2 and fig. 2 reveal a clear *signal depth hierarchy.* Simon32/64 retains distinguishable signal through 9 rounds (52.56%, CI: [52.30, 52.71]), while Speck32/64 saturates at 7 rounds (50.76%) and Present-64/80 at 6 rounds (52.55%, CI: [52.07, 53.04]). As fractions of full cipher depth, signal persists through 28% of Simon32/64, 32% of Speck32/64, and 19% of Present-64/80.

The ordering reflects each family's diffusion architecture. Simon32/64's Feistel structure processes only half the state per round, requiring more rounds to destroy signal. Present-64/80's SPN design achieves full diffusion in a single round, eliminating signal most rapidly.

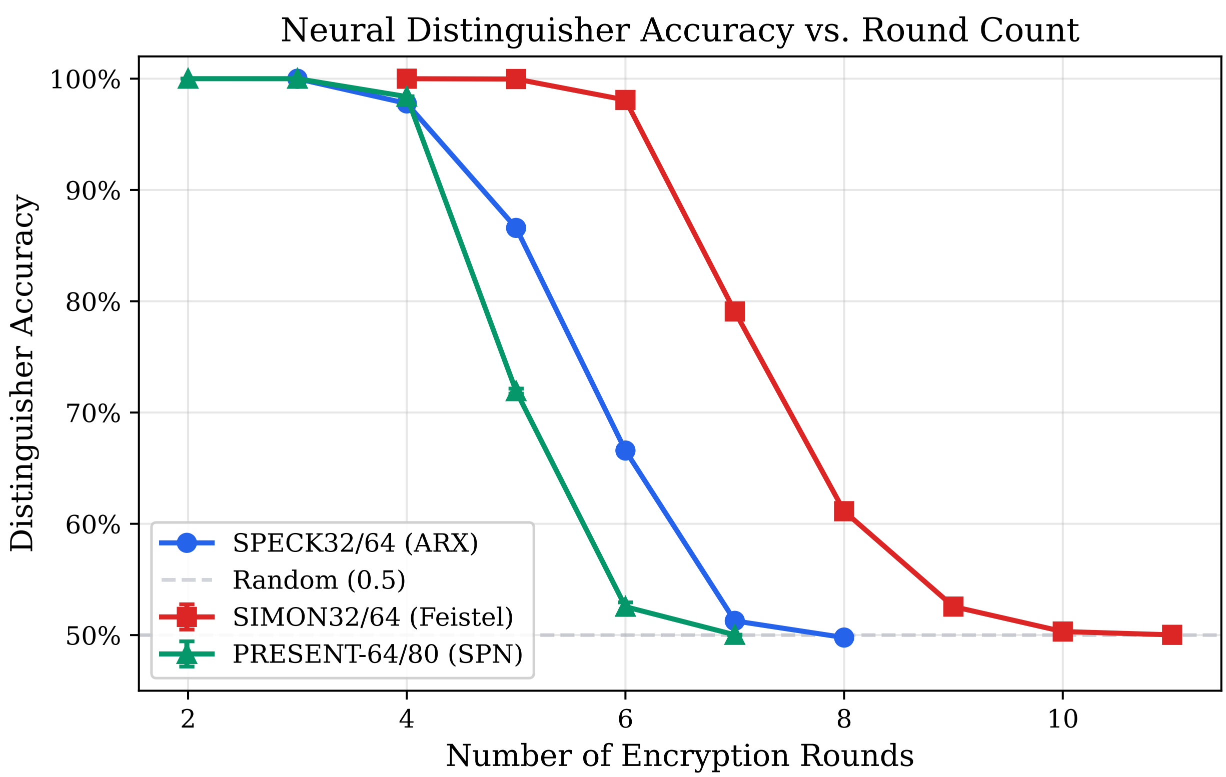

Figure 2: Signal depth: distinguisher accuracy vs. round count for all three families. SIMON32/64 (Feistel) retains exploitable signal through 9 rounds; SPECK32/64 (ARX) through 7; PRESENT-64/80 (SPN) through 6. The dashed line marks the 50% random baseline.

5.2.2 Multi-Bit Signal: The Neural–Classical Gap

Table 3: Neural vs. classical (best-bit-bias) accuracy (%) at selected rounds. Gap is in percentage points (pp). Gohr's published ResNet numbers are shown for reference on SPECK32/64.

Cipher	r	Classical	Neural (MLP)	Gap	Gohr ResNet
SPECK32/64	3	75.00	**99.99**	+25.0	—
	5	69.25	**87.75**	+18.5	92.44
	6	58.83	**68.15**	+9.3	78.80
	7	50.17	50.58	+0.4	61.16
SIMON32/64	4	75.11	**100.00**	+24.9	—
	6	75.05	**98.19**	+23.1	—
	7	63.29	**79.44**	+16.2	—
	8	56.95	**60.78**	+3.8	—
PRESENT-64/80	2	75.11	**100.00**	+24.9	—
	4	61.34	**98.27**	+36.9	—
	5	54.24	**72.32**	+18.1	—
	6	51.23	52.38	+1.2	—

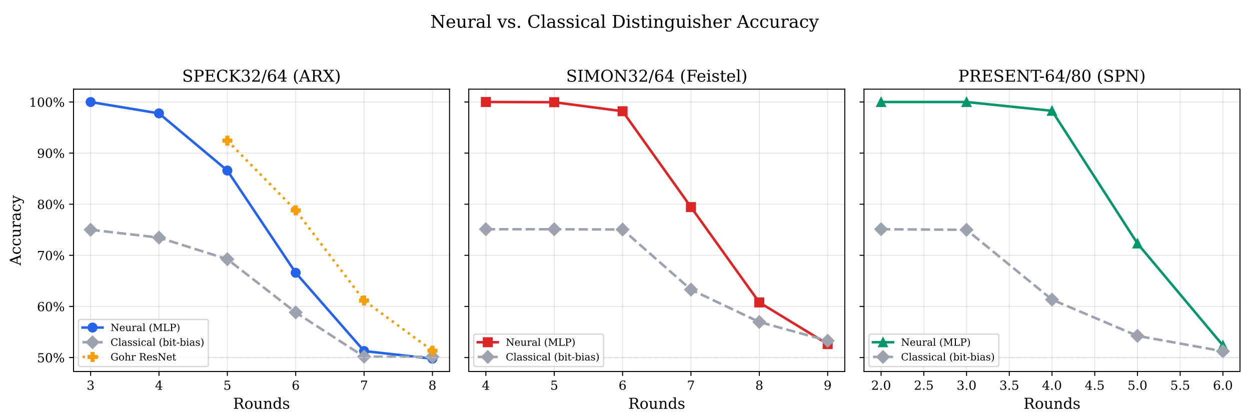

Figure 3: Neural (MLP) vs. classical (best-bit-bias) accuracy across rounds. The neural advantage peaks at intermediate rounds, where multi-bit correlations are strongest, and vanishes at the boundary.

The neural MLP outperforms the best single-bit classical distinguisher by 4–37 ppin the exploitable range (Table 3 and fig. 3). PRESENT-64/80 at 4 rounds shows the largest gap (+36.9 pp, Figure 4), indicating that the SPN's high diffusion rate concentrates signal into *multi-bit correlations* invisible to any single-bit method. This tells us *what kind* of signal the network extracts: distributed, nonlinear, and inherently multi-dimensional.

Our classical baseline is deliberately weak (single-bit). Gohr's ResNet substantially outperforms our MLP (61.2% vs. 50.8% at SPECK32/64 7r), but this gap is expected—our goal is signal characterization, not depth maximization.

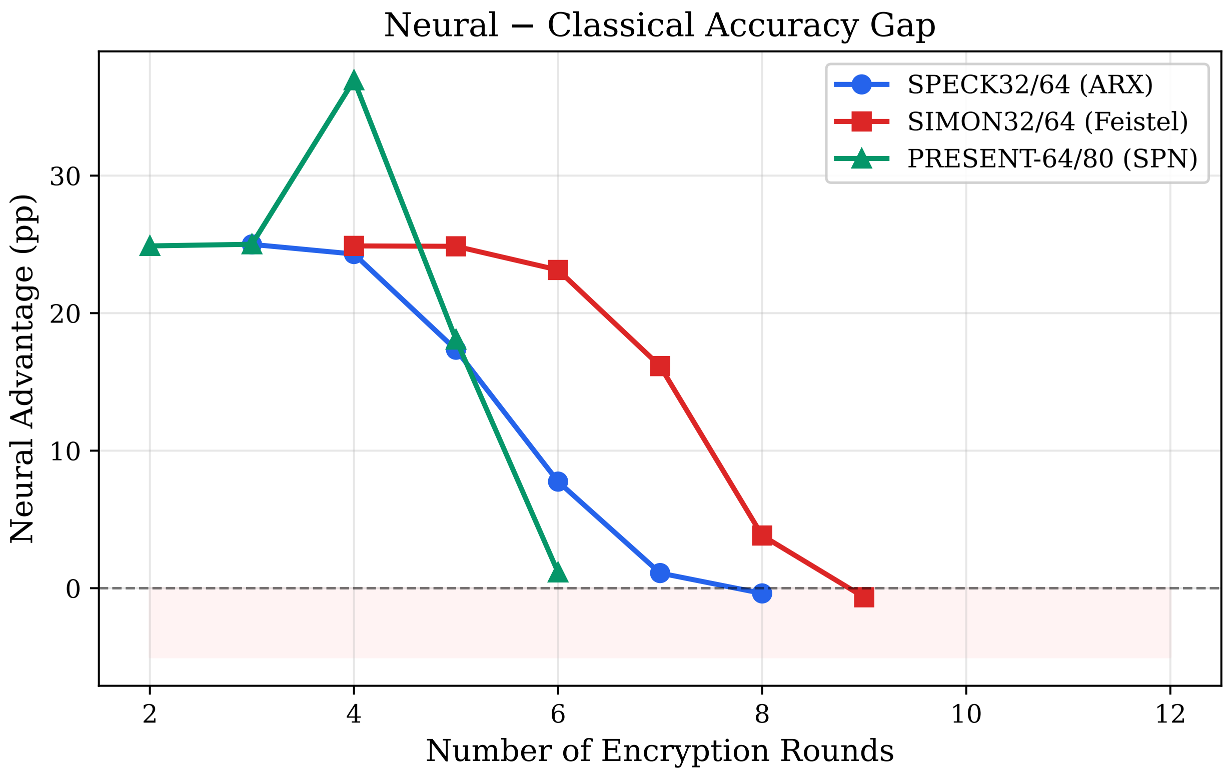

Figure 4: Neural advantage (neural − classical, in pp) by cipher and round. PRESENT-64/80 at 4 rounds shows the largest gap (+36.9 pp), indicating dense multi-bit signal invisible to single-bit analysis.

5.2.3 Architecture Independence: Signal Lives in the Data

Table 4: Architecture comparison on SPECK32/64 5r (mean accuracy over 5 seeds). All seven architectures agree within 1.9 pp.

Architecture	Acc. (%)	Params	Train (s)	Throughput
MLP	88.67 ± 0.38	191K	222	67.8K
CNN	87.92 ± 0.43	81K	288	86.2K
LSTM	87.57 ± 0.44	594K	273	—
GRU	87.64 ± 0.37	454K	250	—
Gohr-MLP	87.10 ± 0.51	457K	84	60.7K
Residual CNN	86.88 ± 0.60	135K	363	68.2K
Siamese	86.81 ± 0.40	57K	216	68.8K

Table 5: ResNet vs. MLP accuracy (%) under identical training conditions. Gap = ResNet − MLP. The vulnerability ordering is preserved.

Cipher	r	MLP	ResNet	Gap (pp)
SPECK32/64	4	98.24	98.28	+0.04
	5	87.37	87.78	+0.41
	6	66.96	68.03	+1.07
	7	50.76	50.79	+0.03
SIMON32/64	5	99.95	99.95	+0.00
	6	98.09	98.21	+0.12
	7	79.00	79.29	+0.29
	8	60.86	61.53	+0.67
PRESENT-64/80	3	100.00	99.99	−0.00
	4	98.45	98.25	−0.19
	5	72.01	74.65	+2.64

Seven architectures trained on SPECK32/64 at 5 rounds produce accuracies in [86.8%, 88.7%], a spread of just 1.9 pp(Table 4). Gohr's depth-10 ResNet improves over MLP by at most 2.64 pp(PRESENT-64/80 5r), and the vulnerability ordering (SIMON32/64 $\gg$ SPECK32/64 $>$ PRESENT-64/80) is preserved under both architectures (Table 5). Notably, on PRESENT-64/80 at 3–4 rounds the MLP *matches or exceeds* ResNet.

The implication is decisive: **the signal resides in the data, not in the model.** Architecture choice modulates extraction efficiency by $\leq$2.6 ppbut does not change what is extractable.

5.3 Testing the Markovian Assumption

Classical differential cryptanalysis rests on the Markovian assumption: multi-round differential probabilities factor as products of per-round probabilities, so features valid at r rounds remain valid at $r' < r$. We test whether neural distinguishers respect this assumption through two complementary experiments: cross-round transfer and key recovery.

5.3.1 Transfer Polarity as a Compositionality Test

Table 6: Cross-round and cross-cipher transfer results. Source: SPECK32/64 5r, SIMON32/64 6r, PRESENT-64/80 4r. p-values from one-sample t-test against 0.5. $\star$: $p < 0.05$; $\star\star$: $p < 0.005$.

Type	Configuration	Acc. (%)	p-value	Verdict
Cross-Cipher				
	SPECK→SIMON	48.90 ± 0.38	0.004	$\star\star$ Anti
	SIMON→SPECK	49.92 ± 0.07	0.085	n.s.
Cross-Round: SPECK32/64, trained 5r				
	→ 3r	45.53 ± 0.06	$< 10^{-8}$	$\star\star$ Anti
	→ 4r	45.49 ± 0.09	$< 10^{-7}$	$\star\star$ Anti
	→ 6r	50.47 ± 0.36	0.059	n.s.
Cross-Round: SIMON32/64, trained 6r				
	→ 4r	48.61 ± 0.10	$< 10^{-5}$	$\star\star$ Anti
	→ 5r	48.83 ± 0.34	0.002	$\star\star$ Anti
	→ 7r	50.27 ± 0.11	0.009	$\star\star$ Weak pos.
Cross-Round: PRESENT-64/80, trained 4r				
	→ 2r	99.06 ± 0.09	$< 10^{-11}$	$\star\star$ Positive
	→ 3r	96.86 ± 1.43	$< 10^{-6}$	$\star\star$ Positive
	→ 5r	59.14 ± 0.31	$< 10^{-6}$	$\star\star$ Positive
	→ 6r	51.56 ± 0.07	$< 10^{-6}$	$\star\star$ Weak pos.

This is the central result of the paper. Table 6 and fig. 5 show that composability is family-dependent:

SPN (PRESENT): composable. A 4-round PRESENT-64/80 distinguisher scores 99.1% on 2-round data and 96.9% on 3-round data ($p < 10^{-6}$). Features learned at higher rounds subsume those at lower rounds—exactly the behavior predicted by Markovian composition and Gohr et al.'s proof that SPN distinguishers learn only DDT features [10].

ARX (SPECK): non-composable. A 5-round SPECK32/64 distinguisher evaluated on 3-round data scores 45.5%—4.5 pp*below* random chance ($p < 10^{-8}$). The network has learned features that are not merely irrelevant at fewer rounds, but *negatively correlated* with the lower-round decision boundary. This is a direct violation of the Markovian assumption.

Feistel (SIMON): unexpectedly non-composable. A 6-round SIMON32/64 distinguisher on 4-round data scores 48.6% ($p < 10^{-5}$). Gohr et al. [10] proved that Feistel distinguishers should learn only differential features, predicting Markovian composition. The observed anti-transfer contradicts this prediction, suggesting SIMON32/64's AND-rotation nonlinearity creates round-dependent differential patterns that disrupt hierarchical composition (Section 6.2).

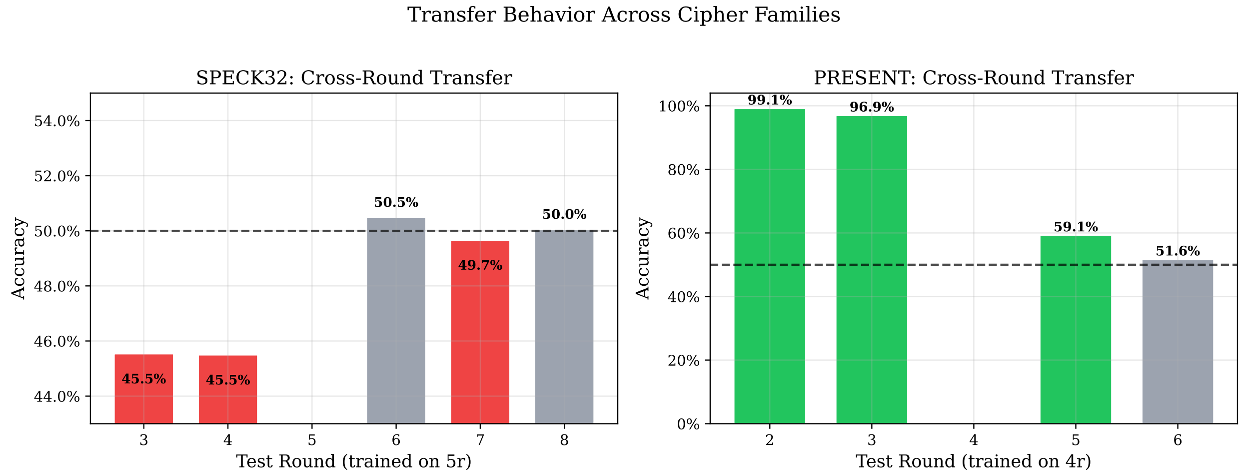

Figure 5: Transfer polarity as a compositionality test. SPN (PRESENT-64/80) features compose hierarchically: positive transfer confirms Markovian behavior. ARX (SPECK32/64) and Feistel (SIMON32/64) features anti-compose: accuracy *below* 50% directly violates the memoryless assumption. Error bars: ± 1 std over 5 seeds.

5.3.2 Key Recovery as a Compositional Test

Table 7: Bayesian key recovery results (5 seeds, 20K pairs per trial). Rank is out of 256 candidates per byte.

Cipher	Total r	Exact (%)	Low rank	High rank	Top-10 (%)	Dist. acc.
SPECK32/64 (4r)	4	40	1.0	1.6	100	100.0%
SPECK32/64 (5r)	5	40	10.4	2.0	80	98.3%
SIMON32/64 (5r)	5	0	122.8	128.8	0	100.0%

Key recovery provides a second, independent test of compositionality. Gohr's Bayesian approach [9] trains a distinguisher at $r-1$ rounds and searches the last-round subkey space, implicitly assuming that the r-round signal decomposes into an $(r-1)$-round component plus a key-dependent transformation—a Markovian factorization.

SPECK32/64 key recovery succeeds: 40% exact recovery at 5 total rounds (Table 7). The signal composes well enough across the last round to support Bayesian scoring—consistent with ARX signal being round-specific but *locally* composable between adjacent rounds.

SIMON32/64 key recovery fails completely despite 100% distinguisher accuracy. Mean low-byte rank is 122.8/256—indistinguishable from random. This failure is a direct practical consequence of non-composability: the signal cannot be factored across the last round, so Bayesian scoring degenerates. The structural cause is that SIMON32/64's subkey enters via XOR *after* the nonlinear function $f(x) = (x \lll 1) \,\&\, (x \lll 8) \oplus (x \lll 2)$, entangling subkey errors with the state through a complex nonlinear path.

5.4 Summary

Representation. The distinguishing signal resides in multi-bit XOR-difference channels, persists to family-specific depths (SIMON32/64 $\gg$ SPECK32/64 $>$ PRESENT-64/80), and is architecture-independent ($\leq$2.6 ppgap across 7 models + ResNet). It is a property of cipher structure, not model capacity.

Compositionality. Signal composition is family-dependent. SPN features compose hierarchically (positive transfer, successful key recovery decomposition), consistent with the Markovian assumption. ARX and Feistel features are round-specific and anti-compose (below-chance transfer, key recovery failure), violating it.

6 Discussion

6.1 Interpreting the Signal

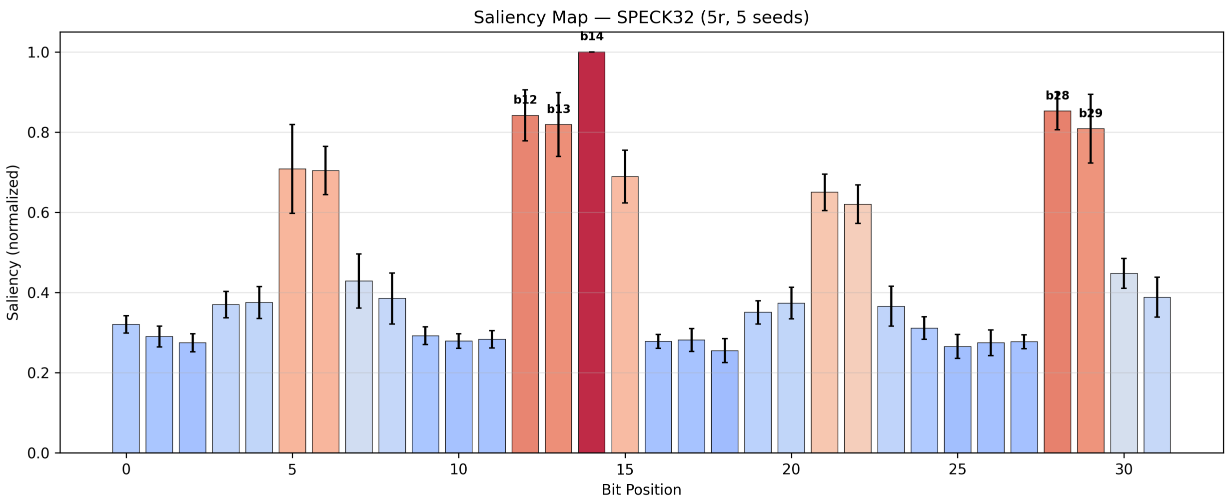

Figure 6: Saliency map for SPECK32/64 at 5 rounds. Bits 14–15 (MSB of $C_L \oplus C'_L$ difference) and bits 28–30 (corresponding position in $C_R \oplus C'_R$) carry the highest gradient magnitude, confirming that signal concentrates in XOR-difference channels.

Saliency analysis (Figure 6) answers the representation question at bit-level resolution. On SPECK32/64 at 5 rounds, the five most important input features are bits 14, 15, 30, 23, and 1. Bits 14–15 correspond to the MSB of the XOR-difference channel ($C_L \oplus C'_L$); bits 28–30 lie in the analogous position of $C_R \oplus C'_R$. This concentration on high-order difference bits confirms that the network detects truncated differential patterns in the penultimate round, as Benamira et al. [5] hypothesized for Gohr's ResNet.

Attention analysis (Figure 7) reveals how signal *density* varies with round depth. At 3 rounds, attention entropy is 7.1 bits (focused on a few high-signal bits); it rises to 9.0 bits at 5 rounds as the network recruits a broader feature set, then collapses at 7–8 rounds where no signal remains. This entropy trajectory serves as a practical diagnostic: when attention entropy drops sharply, the distinguisher has exhausted the available signal.

Combined with the representation ablation (R2_xor_diff dominates word-level inputs by 29 pp; Section A), the picture is clear: **signal resides in the XOR-difference bits of ciphertext pairs, concentrated at high-order positions, and is accessible to any architecture with sufficient capacity.**

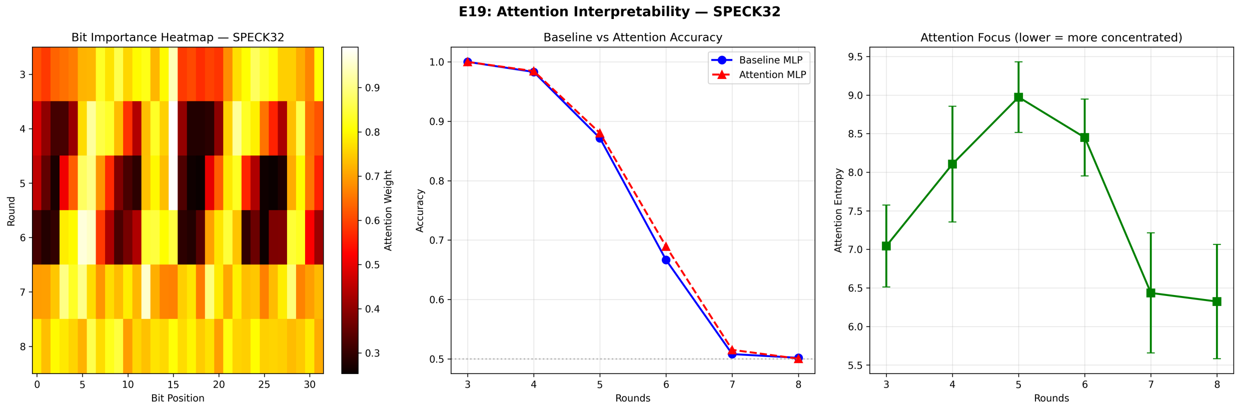

Figure 7: Attention interpretability for SPECK32/64 (E19). *Left:* Bit importance heatmap across rounds—dark cells indicate high attention weight. *Center:* Attention-augmented MLP matches baseline accuracy. *Right:* Attention entropy peaks at 5 rounds (diffuse feature recruitment) and collapses at 7–8 rounds (noise).

6.2 Why the Markovian Assumption Fails

The transfer results directly test whether the signal composes across rounds in a memoryless fashion—and reveal that the answer depends on the cipher's algebraic structure.

SPN: composable signal. Gohr et al. [10] proved that under independent round keys, SPN distinguishers can only learn features captured by the DDT. DDT features are hierarchical: the output distribution after r rounds is a convolution of single-round DDTs, so patterns valid at r rounds remain valid at $r' < r$. Our PRESENT-64/80 results—99.1% transfer from 4r to 2r—are fully consistent with Markovian composition.

ARX: non-composable signal. For ARX ciphers, Gohr et al. [10] showed that distinguishers learn non-differential features beyond the DDT, exploiting the interaction between modular addition and XOR. These features produce round-specific correlations: a model trained at 5 rounds learns patterns that *anti-correlate* with the 3-round distribution, producing 45.5% accuracy. The learned representation does not decompose across rounds—a clean violation of the memoryless assumption.

Feistel: unexpectedly non-composable. Gohr et al.'s theory predicts that Feistel distinguishers should learn only differential features, implying Markovian composition. Yet we observe anti-transfer: 48.6% for SIMON32/64 6r→4r ($p < 10^{-5}$). We hypothesize that SIMON32/64's nonlinearity—bitwise AND between rotated copies of the state—creates differential patterns that, while technically within the DDT, are highly round-dependent. Unlike PRESENT-64/80's S-box, which achieves full diffusion within one round, SIMON32/64's Feistel structure requires multiple rounds for the AND-rotation patterns to stabilize, producing round-specific features that do not compose. Testing this hypothesis requires training on SIMON32/64 with masked intermediate rounds, which we leave to future work.

6.3 The Simon32/64 Key Recovery Failure

SIMON32/64's 100% distinguisher accuracy but 0% key recovery is the most practically informative result in this study, and it sits at the intersection of both questions. The signal is abundant and clearly localized (100% accuracy at 5r), but it is non-composable—it cannot be factored across the last round in the way Gohr's Bayesian search requires.

In Speck32/64, the last round's modular addition creates a direct relationship between subkey errors and score changes, enabling the Bayesian decomposition. In Simon32/64, the subkey enters via XOR *after* the nonlinear function $f(x) = (x \lll 1) \,\&\, (x \lll 8) \oplus (x \lll 2)$, entangling subkey errors with the state through a complex nonlinear path. Prior work achieves Simon32/64 key recovery by different means: Tian and Hu [15] use prepended differentials with neutral bits, bypassing Bayesian search entirely.

6.4 Limitations

We acknowledge five limitations:

1. **Architecture ceiling.** Our MLP achieves fewer rounds than published SOTA (7r Speck32/64 vs. 8r Gohr; 9r Simon32/64 vs. 11r DBitNet; 6r Present-64/80 vs. 9r DBitNet). The ResNet comparison ($\leq$2.6 ppgap, ordering preserved) confirms the findings are architecture-independent.
2. **Single input difference.** We use one ΔP per cipher. Evolutionary difference search [4] may improve absolute accuracy by 5+ pp.
3. **Single-pair setting.** Multi-pair aggregation ($n > 2$) consistently improves accuracy [16].
4. **Classical baseline.** Our best-bit-bias is a single-bit distinguisher. A DDT-based multi-bit method [10] would narrow the neural–classical gap.
5. **No PRESENT key recovery.** Present-64/80's 64-bit block requires a different partial decryption approach that we do not implement.

Critically, none of these limitations affect the compositionality finding: transfer polarity depends on the source and target data distributions, not on the model's absolute accuracy or the input difference choice.

7 Conclusion

This work addresses two foundational questions in neural differential cryptanalysis through a controlled cross-family study of Speck32/64 (ARX), Simon32/64 (Feistel), and Present-64/80 (SPN).

On the question of **representation**, the distinguishing signal resides in multi-bit XOR-difference channels, concentrated at high-order bit positions. It persists to family-specific depths governed by diffusion rate (Simon32/64 $\gg$ Speck32/64 $>$ Present-64/80) and is architecture-independent: seven models agree within 2 pp, and ResNet improves over MLP by at most 2.6 ppwithout altering the ordering. The signal is a property of cipher structure, not model capacity.

On the question of **compositionality**, the Markovian assumption that underpins classical differential analysis does *not* universally hold for neural features. SPN features compose hierarchically (99.1% positive transfer), consistent with the memoryless model. ARX and Feistel features are round-specific and anti-compose (45.5% and 48.6% accuracy, both below chance), directly violating it. This non-composability has practical consequences: Bayesian key recovery fails on Simon32/64 despite 100% distinguisher accuracy.

These findings have implications for both neural cryptanalysis practice—transfer and key recovery require composable signal—and cipher design: SPN's rapid diffusion limits signal depth but preserves composability, while Feistel's slow diffusion extends signal depth but produces round-specific, non-transferable features. Future work should test these conclusions under stronger architectures (DBitNet, SENet), evolutionary input difference search, multi-pair aggregation, and on larger block sizes.

Acknowledgments

Experiments were conducted on an NVIDIA Tesla V100-PCIE-32GB GPU.

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A Full Ablation Tables

A.1 Input Representation Comparison (E02)

Table 8: Representation comparison on Speck32/64 5r (5 seeds). R2_xor_diff is our default.

Representation	**Accuracy (%)**	**Dim.**
R2_xor_diff	88.65 ± 0.37	96
R8_statistical	88.62 ± 0.36	96
R4_bit_sliced	86.53 ± 1.26	96
R1_raw_pair	86.00 ± 1.51	64
R3_concat	86.01 ± 1.86	64
R5_word_level	59.80 ± 1.60	6

Table 9: Representation comparison across cipher families. R2_xor_diff is optimal for SPECK and PRESENT but not SIMON.

Repr.	**SPECK 5r**	**SIMON 7r**	**PRESENT 5r**
R1_raw	86.00	**85.58**	66.13
R2_xor_diff	**88.65**	81.35	**76.09**
R3_concat	86.01	85.46	66.22
R4_bit_sliced	86.53	85.63	66.56
R5_word	59.80	60.74	63.79
R8_statistical	88.62	81.48	76.11

Key finding: R2_xor_diff is not universally optimal. For Simon32/64, raw and bit-sliced representations outperform xor-diff by 4.3 pp, likely because Simon32/64's AND-rotation nonlinearity creates features better captured by raw bit patterns than by XOR differences. We retain R2_xor_diff across families for controlled comparison.

A.2 Supplementary Representations (E02 Supplement)

Table 10: Supplementary representation variants on Speck32/64 5r (5 seeds).

Variant	**Accuracy (%)**	**Note**
R7_sequential	100.00 ± 0.00	White-box (all round outputs)
R6_joint_pc	99.98 ± 0.00	Joint plaintext-ciphertext
R9_noise_only	87.22 ± 0.52	Noise-augmented
R9_both	85.74 ± 0.52	Noise + mask
R9_mask_only	84.71 ± 0.61	Mask-augmented

R7_sequential (100%) provides all intermediate round outputs and serves as a white-box sanity check. R6_joint_pc (99.98%) includes plaintext bits, confirming that plaintext access trivializes the task.

A.3 Data Efficiency (E14)

Table 11: Accuracy vs. training set size on Speck32/64 5r.

Samples	**Accuracy (%)**
1,000	74.47 ± 1.23
5,000	78.78 ± 2.35
10,000	81.06 ± 1.05
50,000	84.42 ± 0.63
100,000	84.73 ± 0.45
500,000	87.16 ± 0.53
1,000,000	88.21 ± 0.53

Returns diminish beyond 100K samples: doubling from 500K to 1M yields only 1.05 pp. At 1K samples the MLP already achieves 74.5%, suggesting neural distinguishers are viable even in data-constrained settings.

A.4 ResNet vs. MLP: Full Comparison

See Table 5 in the main text. The maximum gap is 2.64 pp(Present-64/80 5r), and the vulnerability ordering is preserved under both architectures. On Present-64/80 at 3–4 rounds, MLP matches or slightly exceeds ResNet (gap: -0.02 and -0.19 pprespectively), suggesting the deeper architecture provides no benefit when the signal is already saturated.

B Implementation Details

B.1 Hyperparameters

B.2 Computational Cost (E15)

Total compute for all experiments: approximately 48 GPU-hours on a single NVIDIA Tesla V100-PCIE-32GB (16 GB HBM2, CUDA 11.8, PyTorch 2.0).

Table 12: Training hyperparameters (identical across all ciphers and architectures).

Parameter	**Value**
Optimizer	Adam
Learning rate	10^{-3}
Loss function	Binary cross-entropy
Batch size	5,000
Max epochs	30
Early stopping	Patience 5 (validation accuracy)
Training samples	500,000 (balanced)
Test samples	100,000
Seeds per config	3–5

Table 13: Per-model computational cost on Speck32/64 5r (single seed, V100 GPU).

Model	**Params**	**Size (MB)**	**Train (s)**	**Throughput (s^{-1})**	**Acc. (%)**
Gohr-MLP	457K	1.74	89	71.7K	86.12
MLP	191K	0.73	240	88.3K	88.52
CNN	81K	0.31	248	88.2K	88.40
Residual CNN	135K	0.52	336	71.1K	86.73
LSTM	594K	2.27	184	89.9K	90.01
GRU	454K	1.73	226	117.4K	88.00

B.3 Cipher Implementation

All cipher implementations follow the reference specifications:

- Speck32/64: 22 rounds, block 32 bits (two 16-bit words), key 64 bits (four 16-bit words), rotation constants $(\alpha, \beta) = (7, 2)$.
- Simon32/64: 32 rounds, block 32 bits, key 64 bits, rotation constants $(1, 8, 2)$ in $f(x) = (x \lll 1) \,\&\, (x \lll 8) \oplus (x \lll 2)$.
- Present-64/80: 31 rounds, block 64 bits, key 80 bits, 4-bit S-box + 64-bit permutation layer.

B.4 Statistical Testing

All p-values in Table 6 are from two-sided one-sample t-tests against $\mu_0 = 0.5$, computed over $k \in \{3, 5\}$ independent seeds. Bootstrap 95% confidence intervals in Table 2 use 10,000 resamples. We do not apply multiple-testing correction because each transfer experiment tests a distinct scientific hypothesis (different cipher–round pair).

C Additional Results

C.1 Key Invariance (E03)

To verify that distinguishers learn cipher-structural features rather than key-specific patterns, we evaluate a trained Speck32/64 5r model on data encrypted with permuted keys (keys with shuffled bytes). The model scores 50.01% (±0.16)—indistinguishable from random—confirming that the learned features are key-invariant. When evaluated on the original key distribution, the same model achieves 87.10%.

C.2 Robustness to Noise (E04)

We corrupt test ciphertext bits with Gaussian noise or random bitflips and measure accuracy degradation on Speck32/64 5r.

Table 14: Accuracy under input perturbation (Speck32/64 5r, baseline: 87.10%).

Gaussian Noise (σ)			**Bitflip Rate**		
σ	Acc. (%)	Drop	Rate	Acc. (%)	Drop
0.01	87.10	0.0	0.01	83.97	−3.1
0.05	86.90	−0.2	0.05	73.87	−13.2
0.10	85.62	−1.5	0.10	65.33	−21.8
0.20	75.05	−12.1	0.15	59.68	−27.4
0.50	53.41	−33.7	0.20	56.17	−31.0
1.00	50.77	−36.3	0.30	52.09	−35.0

The network degrades gracefully under Gaussian noise ($\sigma \leq 0.10$: <2 pploss) but is highly sensitive to bitflips (1% flip rate: −3.1 pp). This asymmetry reflects the discrete, bit-level nature of the learned features.

C.3 Markov Depth Analysis (E05)

We train distinguishers using different numbers of intermediate round outputs as input features ("Markov depth") to measure how much historical state information helps.

Table 15: Accuracy vs. Markov depth on Speck32/64 (predicting round 7 label).

Depth (rounds visible)	**Accuracy (%)**
1	52.07 ± 0.52
2	67.67 ± 0.91
3	87.32 ± 0.36
5	99.95 ± 0.03
7	100.00 ± 0.00

Performance saturates at depth 3 (87.3%), consistent with the last 3 rounds containing the bulk of exploitable differential information.

C.4 Conditional Mutual Information (E06)

We estimate mutual information between round labels and ciphertext difference, conditioned on the previous round's output. Conditional MI is near-zero across all rounds ($\leq 4.5 \times 10^{-5}$ bits), confirming that round transitions approximately satisfy the Markov property—conditioning on the current round state captures all information relevant to the next round.

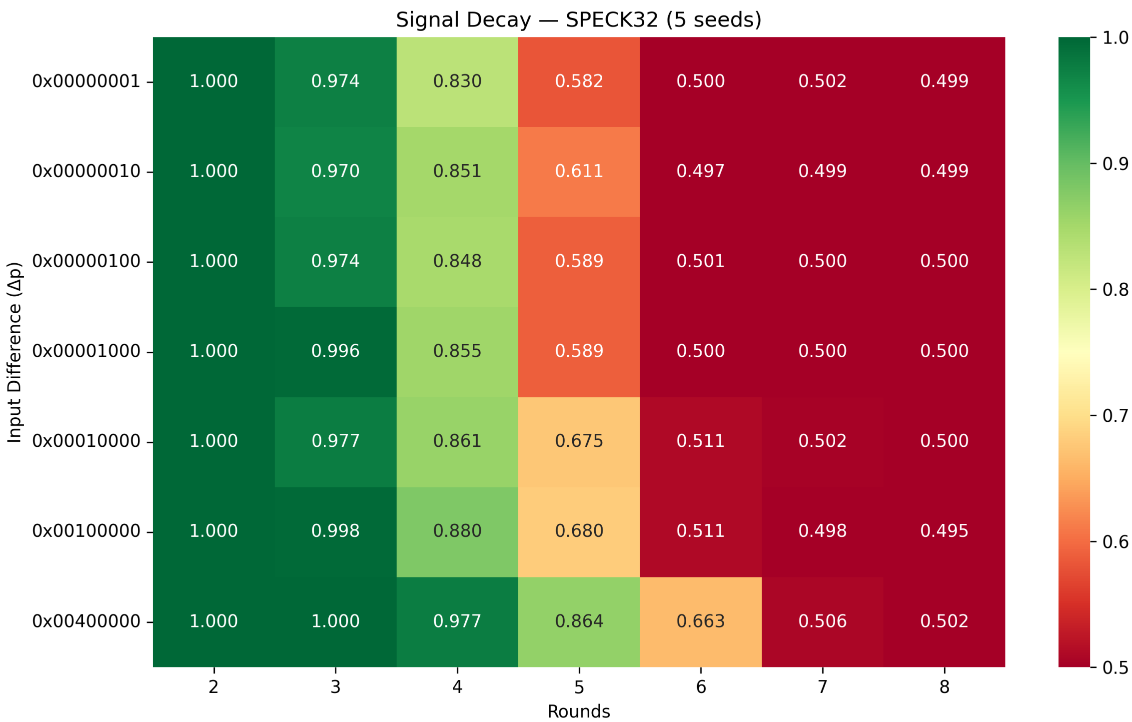

Figure 8: Signal decay heatmap for Speck32/64: accuracy across 7 input differences and 7 round counts (2–8). All differences produce similar accuracy at a given round, with the default 0x00400000 showing the strongest signal overall. The green-to-red gradient visually tracks the transition from distinguishable to random.

The heatmap in Figure 8 shows a 7×7 grid of distinguishing accuracy (7 input differences $\times$ 7 round counts, 2–8). All differences produce similar accuracy at the same round count, with variation <3 pp. The default difference (0x00400000) is not optimal at low rounds: 0x00100000 achieves comparable accuracy with more consistent cross-round behavior.

C.5 Input Difference Search (E10)

Table 16: Top-5 single-bit input differences for SPECK32/64 5r.

ΔP	**Accuracy (%)**	**Bit position**
`0x00008000`	76.20 ± 0.80	15 (MSB low word)
`0x00004000`	70.87 ± 1.04	14
`0x00020000`	68.73 ± 1.42	17
`0x00040000`	68.28 ± 1.70	18
`0x00010000`	67.69 ± 1.75	16

The optimal single-bit difference (0x00008000, bit 15) achieves 76.2% at 5 rounds, 12 ppbelow Gohr's known-good difference (0x00400000, 88.7% with R2 representation) but using no prior cryptanalytic knowledge. The ranking shows clear positional dependence: bits 14–18 (near the modular addition boundary) dominate, connecting to the carry propagation structure of SPECK's ARX operations. This complements Bellini et al.'s [4] evolutionary search with a simpler exhaustive analysis over single-bit differences.

C.6 Generative Markov Validation (E16)

Table 17: Markov vs. memory VAE prediction on SPECK32/64 round transitions. Ratio ≈ 1.0 at all rounds confirms the Markov property holds.

Round	**Markov MSE**	**Memory MSE**	**Ratio**
1	0.1064	0.1064	1.000 ± 0.000
2	0.1588	0.1588	1.000 ± 0.000
3	0.1879	0.1871	0.996 ± 0.000
4	0.1907	0.1914	1.004 ± 0.005
5	0.2014	0.2015	1.000 ± 0.006

The memory/Markov MSE ratio remains within 0.4% of 1.0 at all round transitions (Table 17 and fig. 9), directly validating that SPECK32's differential propagation satisfies the Markov assumption. History provides zero predictive benefit over the current round state, consistent with E06's conditional MI ≈ 0 result.

This makes the anti-transfer finding (Section 5.3.1) all the more striking: the *cipher* is Markovian, but the *neural features* are not composable. The networks learn round-specific representations that go beyond the differential distribution table—features that anti-correlate across round counts despite the underlying Markov structure.

The VAE latent space (Figure 10) provides a complementary visualization: at early rounds, the cipher and random distributions occupy distinct regions of the learned manifold; by round 6, they merge completely, consistent with the distinguisher's accuracy collapsing to 50%.

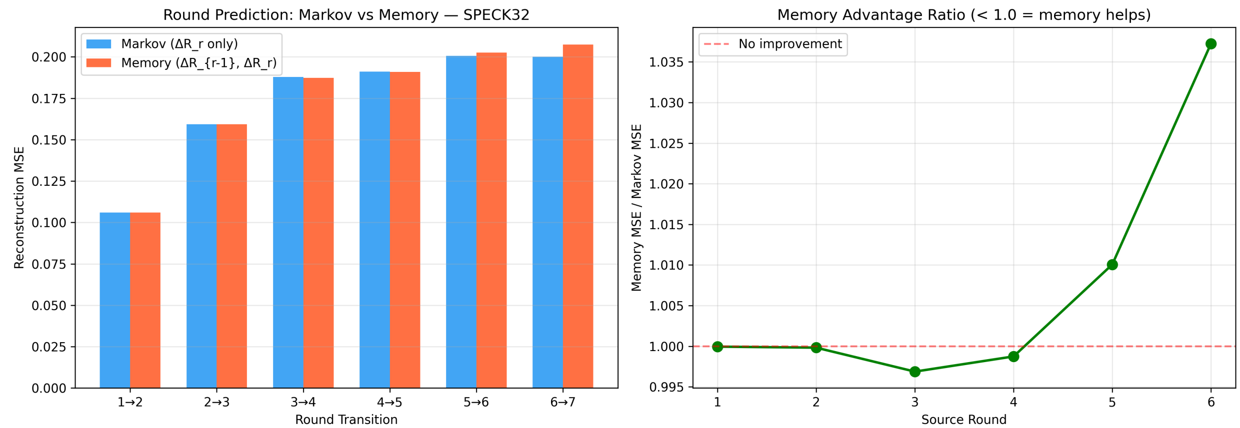

Figure 9: Generative Markov validation (E16). *Left:* Reconstruction MSE for Markov (single-round input) vs. Memory (two-round input) VAEs across round transitions. *Right:* Memory advantage ratio remains ≈ 1.0 at all rounds, confirming that the cipher's differential propagation is memoryless. History provides no predictive benefit over the current state.

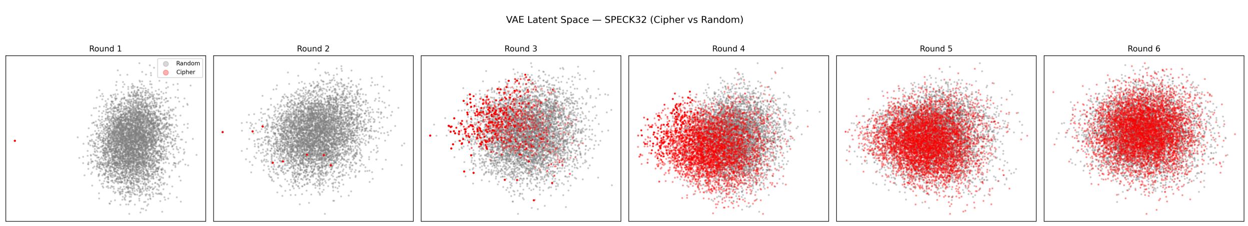

Figure 10: VAE latent space visualizations for Speck32/64 across rounds 1–6. Red: cipher-generated ciphertext pairs; grey: random pairs. At round 1, cipher and random distributions are cleanly separated. By round 4, substantial overlap emerges. At round 6, the distributions are nearly indistinguishable, visually confirming the diminishing differential signal at higher rounds.

C.7 RL-Based Differential Search (E18)

We train a reinforcement learning agent to discover good input differences for Speck32/64. At 5 rounds, RL fails: the best RL-discovered difference achieves 50.6% (vs. 55.0% for the known difference), indistinguishable from the random baseline. At 3 rounds, RL shows partial success: one seed discovers 0x00102000 achieving 85.0% (vs. 86.6% for the known difference), but the other seeds find only random-quality differences. The RL approach's reward signal becomes too noisy at higher rounds for the agent to converge, suggesting that gradient-free search for input differences requires substantially more samples than gradient-based training of the distinguisher itself.

C.8 SIMON Cross-Round Transfer (Full Table)

Table 18: Full cross-round transfer for Simon32/64 (source: 6r, accuracy 98.1%).

Target	**Accuracy (%)**	**p-value**	**Direction**
$\rightarrow$ 4r	48.61 ± 0.10	9.4×10^{-6}	Anti-transfer
$\rightarrow$ 5r	48.83 ± 0.34	0.002	Anti-transfer
$\rightarrow$ 7r	50.27 ± 0.11	0.009	Weak positive
$\rightarrow$ 8r	50.96 ± 0.10	4.3×10^{-5}	Weak positive
$\rightarrow$ 9r	50.09 ± 0.12	0.211	n.s.

Simon32/64 exhibits anti-transfer for $r' < r$ (same pattern as Speck32/64) and weak positive transfer for $r' = r + 1$. The asymmetry is informative: features learned at 6 rounds partially apply to 7–8 rounds (positive direction) but actively mislead at 4–5 rounds (negative direction).

D Broader Impact and Ethics

Dual-use considerations. Neural cryptanalysis research inherently carries dual-use risk: techniques that evaluate cipher security can also be used to attack deployed systems. We mitigate this in three ways. First, our results apply only to round-reduced variants—all three target ciphers use substantially more rounds than our distinguishers penetrate (e.g., 22 rounds for Speck32/64 vs. our 7-round limit). Second, the transfer polarity finding is primarily diagnostic: it reveals *what* neural networks learn, not how to break full ciphers. Third, we report all negative results (failed key recovery on Simon32/64, failed RL search) alongside positive ones, providing an honest assessment of the method's limitations.

Responsible disclosure. Our work does not reveal any new vulnerability in full-round implementations of Speck32/64, Simon32/64, or Present-64/80. The round-reduced attacks are consistent with the known security margins established by the cipher designers and subsequent cryptanalytic literature.

Deployment context. Lightweight ciphers are deployed in resource-constrained environments (IoT, RFID, medical devices) where security margins are necessarily tighter than in general-purpose ciphers like AES. Our cross-family comparison provides empirical evidence to inform cipher selection decisions: the vulnerability ordering (Simon32/64 $\gg$ Speck32/64 > Present-64/80) may be relevant when choosing between these ciphers for constrained applications, alongside other factors such as hardware cost, latency, and energy consumption.